

CROSS-ANATOMY DATA FUSION AND COMPATIBILITY MODELING EMBODIMENTS

In various embodiments, measurements obtained from the intraluminal compliance and distance profiling device may be integrated with measurements obtained from additional sensing devices or user profiles to generate composite models describing interaction compatibility, structural alignment, deformation coordination, and physiologic synchrony between different anatomical systems.

The computational system may combine distance measurements, compliance characteristics, deformation response profiles, stability metrics, and temporal behavior patterns with complementary measurements acquired from external geometry scanners, longitudinal intraluminal monitors, wearable biometric devices, or future sensing nodes. Data fusion may occur locally within a user computing device, within a cloud-based computational environment, or through distributed hybrid processing architectures.

In certain embodiments, cross-anatomy modeling generates derived parameters describing proportional matching, alignment tolerance, interaction stability, predicted comfort response, or mechanical compatibility between interacting anatomical profiles. Such derived metrics may be calculated using statistical correlation models, rule-based algorithms, physics-informed modeling, machine learning systems, or combinations thereof.

The computational system may normalize measurements from different sensing modalities into standardized reference frameworks prior to fusion. Normalization may include dimensional scaling, confidence weighting, calibration compensation, temporal averaging, and conversion into abstracted indices configured to preserve privacy while maintaining analytic meaning.

In some implementations, compatibility modeling operates independently of explicit anatomical labeling. Abstracted structural and compliance parameters may be compared mathematically without exposing sensitive anatomical information, thereby enabling privacy-preserving interaction analysis and anonymized matching workflows.

Data fusion workflows may further support longitudinal adaptation in which compatibility models improve as additional measurement sessions are accumulated. Learning algorithms may refine predictive performance using repeated measurements, anonymized population datasets, or adaptive personalization models evolving over time.

In certain embodiments, the system generates compatibility indicators, alignment scores, interaction profiles, or synchronization metrics derived from combined datasets. Outputs may be presented as numerical scores, graphical indicators, multi-axis visualizations, symbolic representations, or other non-explicit visualization formats intended to maximize interpretability while preserving discretion.

The disclosed architecture further supports interoperability between independently owned devices or profiles. Secure comparison protocols may permit analysis between consenting users while maintaining encryption, anonymization, permission control, and user-governed authorization mechanisms.

In addition, compatibility modeling may inform device personalization, calibration adjustments, guidance workflows, or future device configuration recommendations. Computational feedback loops may therefore enhance both individual measurement accuracy and broader ecosystem intelligence.

Accordingly, cross-anatomy data fusion embodiments extend the disclosed device beyond isolated measurement functionality, enabling coordinated multi-device analysis, adaptive personalization, scalable compatibility computation, and long-term physiologic modeling within an integrated computational ecosystem.